

MODULE-2

TRANSISTORS: BJT AND FET

Bipolar Junction Transistor: BJT Structure, BJT Operation, Characteristics and parameters, BJT as an Amplifier, BJT as a switch. Biasing, Need for Biasing, DC operating point (Q-point).

JFET: JFET Structure, JFET Operation, JFET Characteristics

Construction and Types of BJT

BIPOLAR JUNCTION TRANSISTOR (BJT).

Bipolar : Conduction takes place due to both charge carriers. (electrons & holes). So, This is called bipolar.

Junction : Combination of two diodes is BJT. So, two junctions (CB junction & EB junction formed) between P-type & N-type semiconductors.

Transistor : Transfer Resistor is Transistor.

- ✓ Forward bias applied in input side of transistor.
- ✓ Reverse bias applied in output side of transistor.
- ✓ Hence, low resistance is transformed to high resistance region.

Applications :-

- ✓ It is used as Amplifiers
- ✓ It can be used as Switches.

BJT $\rightarrow$ Controlled output current by input current. So, This device is a current controlled device

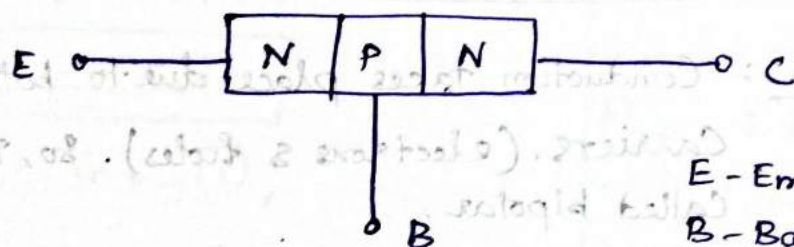
Types of BJT :-

- NPN transistor

- PNP transistor

NPN transistor :-

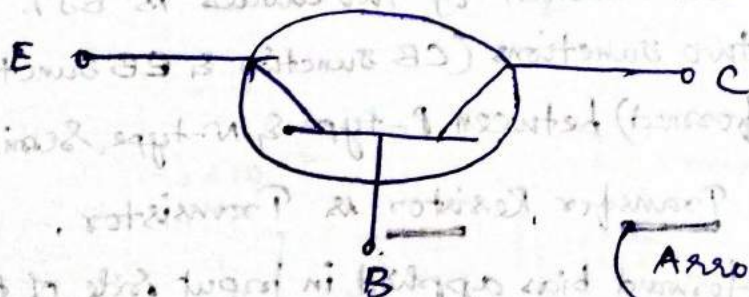
- A thin layer of P-type material is sandwiched between two layers of N-type material.



E - Emitter

B - Base

C - Collector



SYMBOL

(Arrow indicates
Conventional current
flow direction)

Emitter :-

- ✓ It is used to inject the charge carriers, (Electrons) in NPN transistor.

- ✓ It is heavily doped region, but area is lesser than Collector.

Base:

- ✓ It is used to pass all the charge carriers (electrons) from emitter to collector.
- ✓ It is lightly doped.

Collector:

- ✓ It is used to collect all the charge carriers (electrons).
- ✓ It is moderately doped, but larger cross-section area than Emitter.

✓ BJT has two Junctions

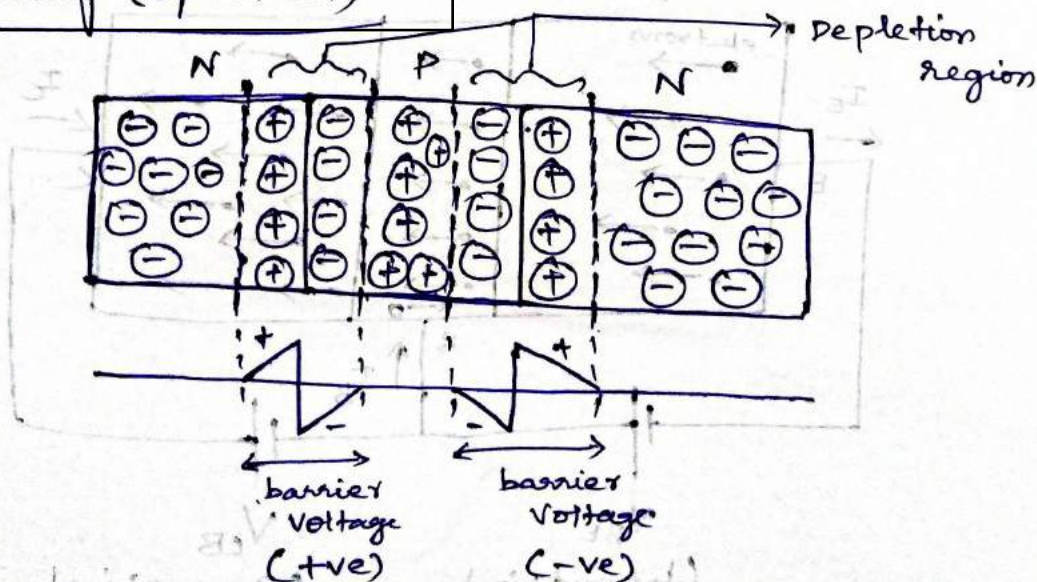
- One junction is between input & ground terminal (Emitter Base junction)
- Another junction is between output & ground terminal (Collector Base junction).

✓ Functions of the Three terminals (Base, Emitter, Collector) are same for all Configurations.

✓ In NPN, Current Conduction is due to electrons.

With a neat diagram explain the operation of an NPN transistor. Also describe the relationship between various currents

Working (Operation):

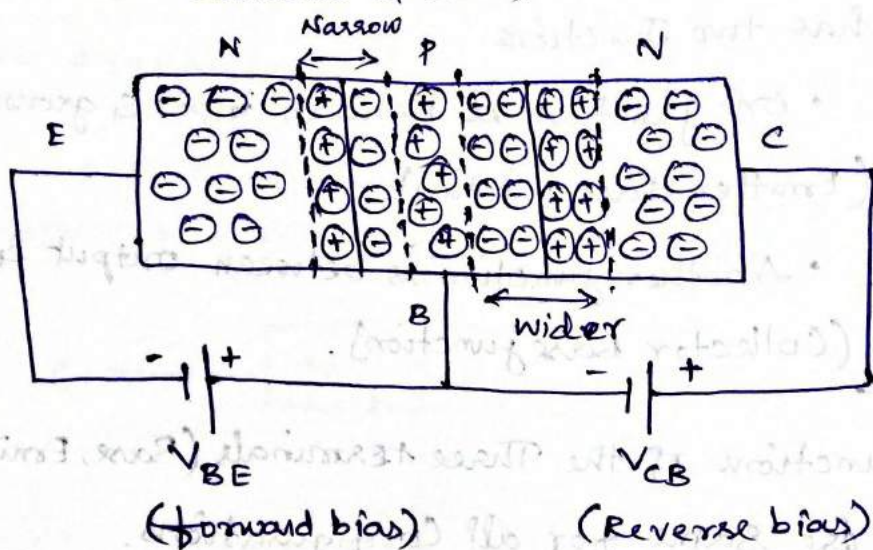


NPN transistor Unbiased Condition

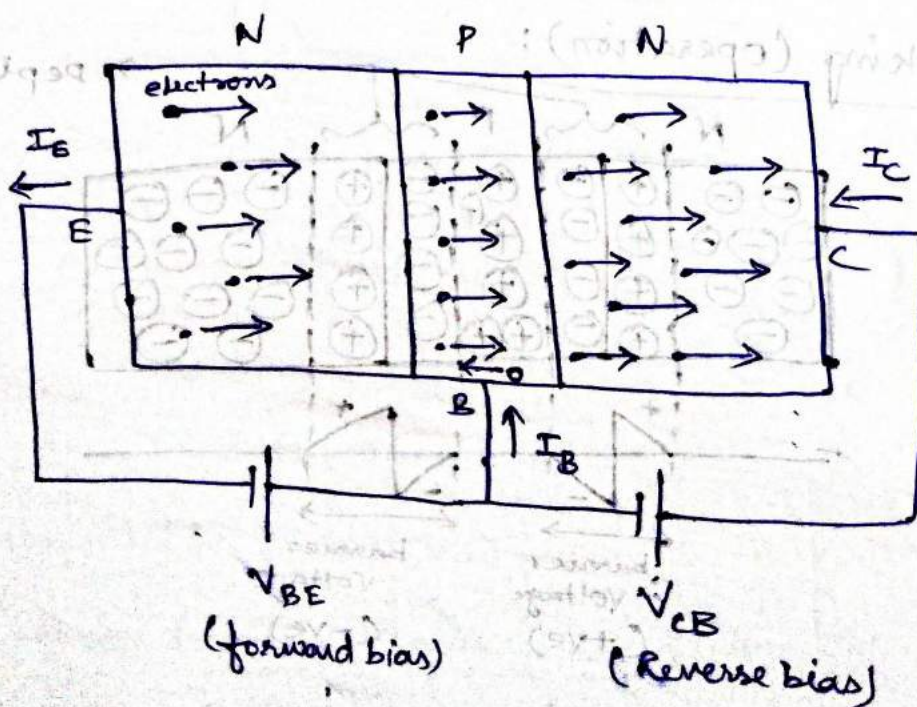
✓ Under Unbiased Condition, the NPN transistor depletion regions & Barrier potentials are shown in figure.

✓ Forward bias is applied to the input side (ie., between emitter & base)

✓ Reverse bias is applied to the output side (ie., between collector & base).



NPN transistor under biased Condition



✓ Due to forward bias, lot of electrons moving from emitter to base & few holes moving from base to emitter. This flow of charge carriers forms emitter current & base current.

$I_E \rightarrow$ Emitter current

$I_B \rightarrow$ Base current

✓ Due to Reverse bias, some of the electrons moving from base to collector & all the electrons moving towards the +ve terminal of reverse bias from collector. This flow of charge carriers forms collector current.

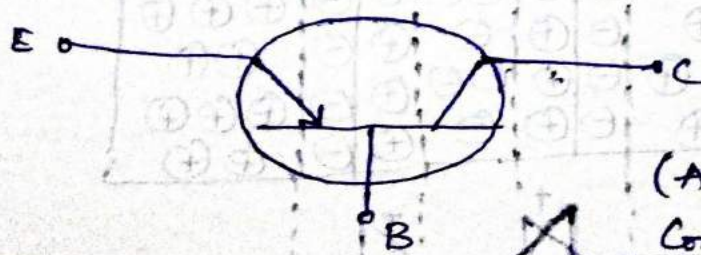
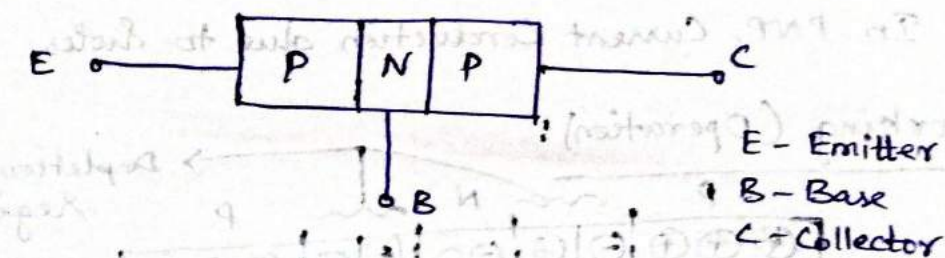
$I_C \rightarrow$ Collector current

✓ Emitter current is sum of Base & Collector current.

$$I_E = I_B + I_C$$

PNP transistor:-

- A thin layer of N-type material is sandwiched between two layers of P-type material.



SYMBOL

(Arrow indicates
Conventional current
flow direction)

Emitter:-

- ✓ It is used to inject the charge carriers (holes) in PNP transistor.
- ✓ It is heavily doped region but area is lesser than collector.

Base:-

- ✓ Base is used to pass all the charge carriers (holes).
- ✓ It is lightly doped.

Collector:-

- ✓ It is used to collect all the charge carriers (holes).
- ✓ It is moderately doped but larger cross section area than emitter.

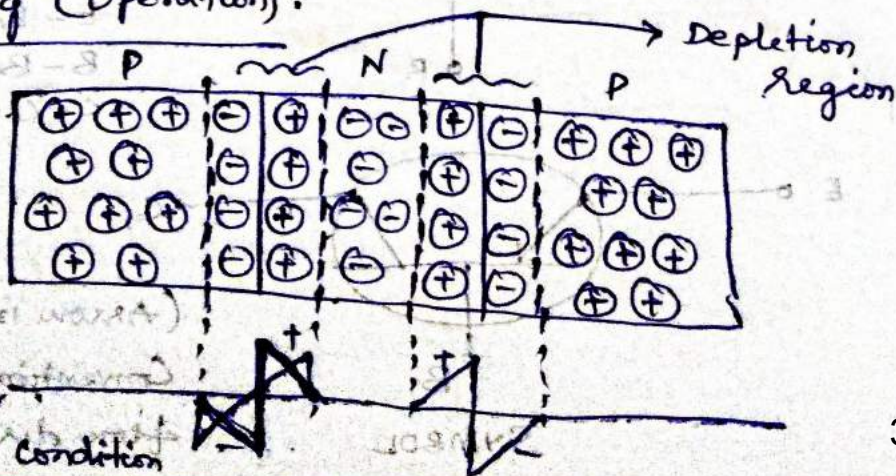
✓ BJT has two junctions

- One junction is between input & ground terminal - (Emitter Base junction)
- Another junction is between output & ground terminal - (Collector Base junction)

✓ Functions of three terminals are same in all configurations.

✓ In PNP, Current conduction due to holes.

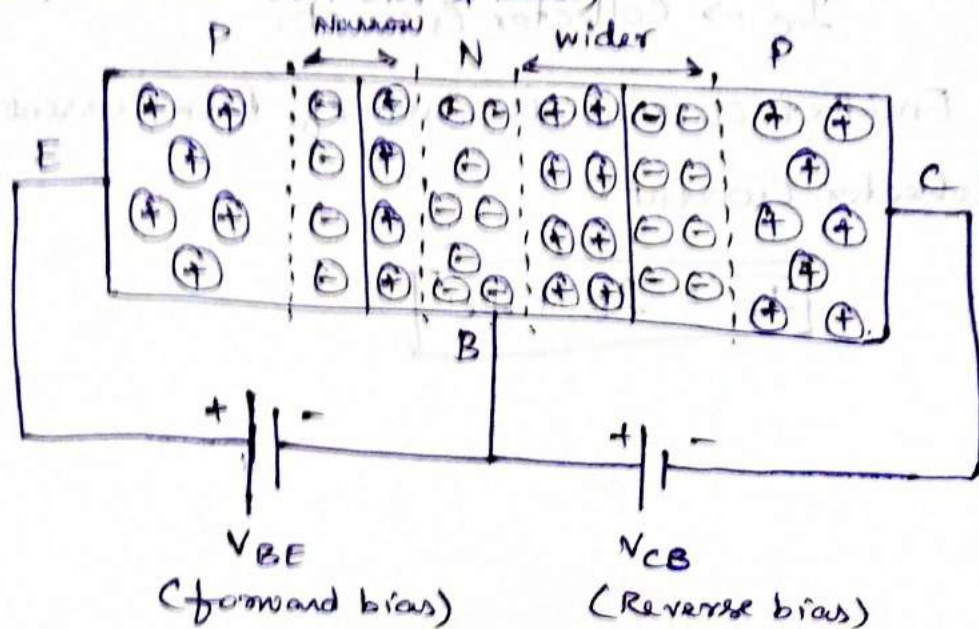
Working (Operation):



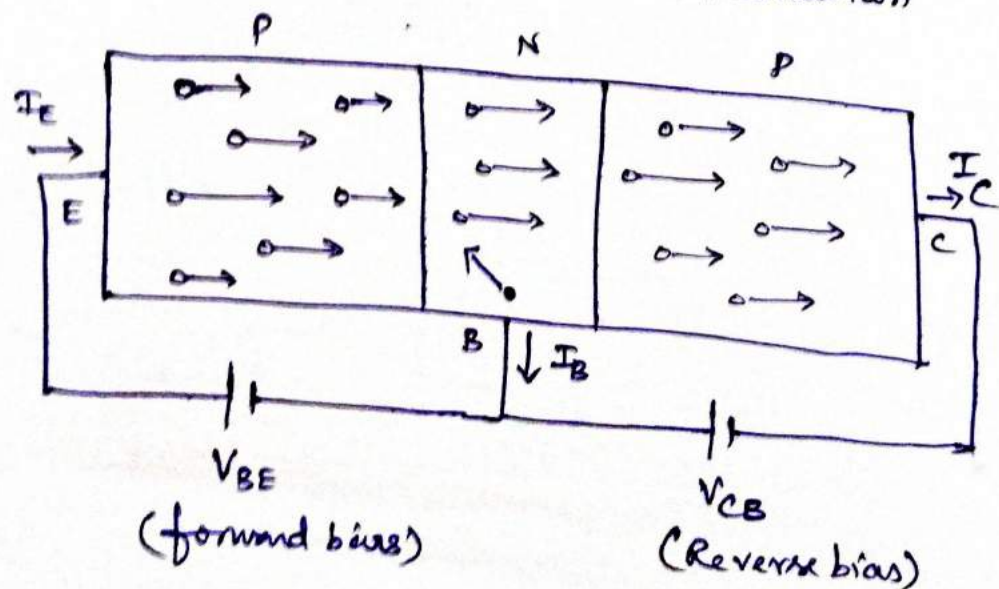
✓ Under Unbiased Condition, the PNP transistor depletion regions & Barrier potentials are shown in figure.

✓ Forward bias is applied to the input side (i.e., between Emitter & Base)

✓ Reverse bias is applied to the output side (i.e., between Collector & Base)



PNP transistor Under biased Condition



✓ Due to forward bias, lot of holes moving from Emitter to base. So, Emitter & base currents formed.

$I_E \rightarrow$ Emitter Current

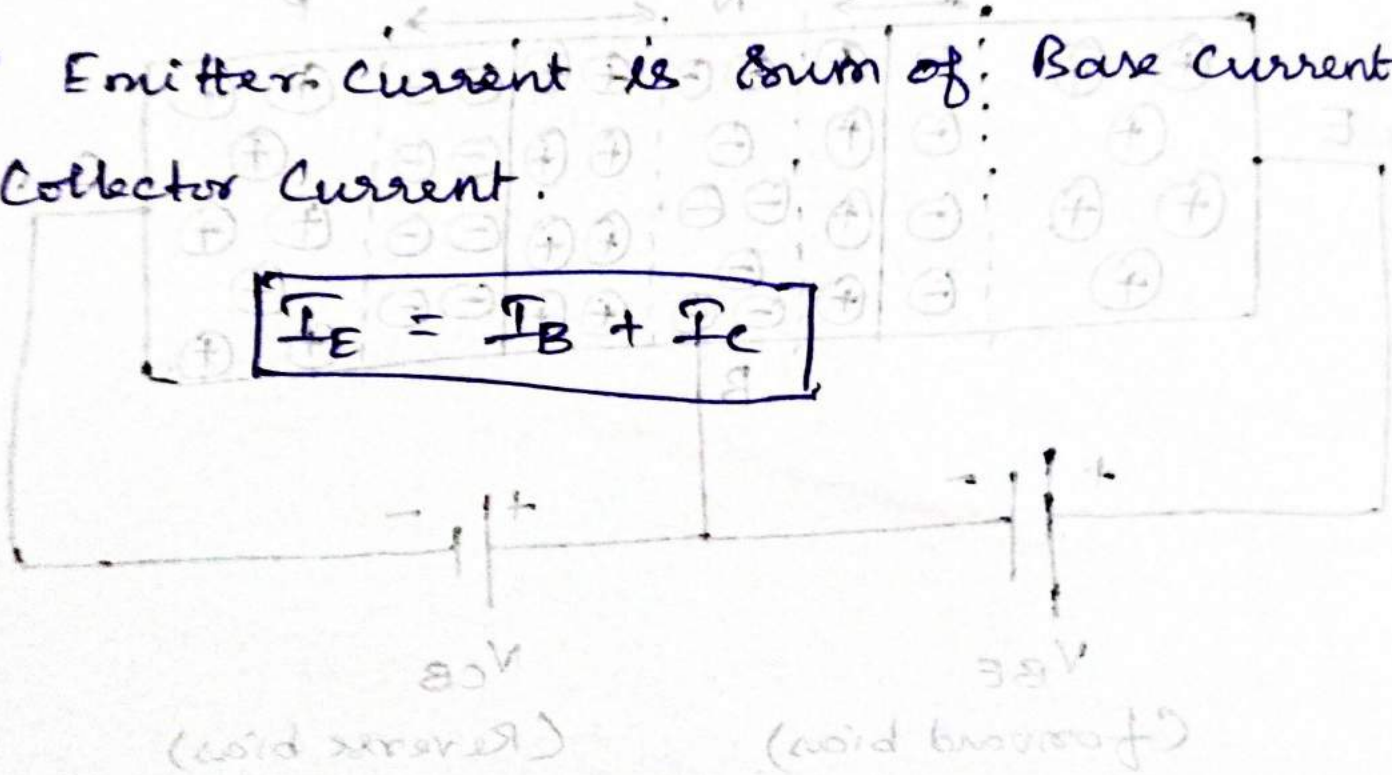
$I_B \rightarrow$ Base Current

✓ Due to Reverse bias, Some of the holes are moving from base to Collector. All the holes are moving towards the -ve terminal of the reverse bias from Collector. This forms Collector Current.

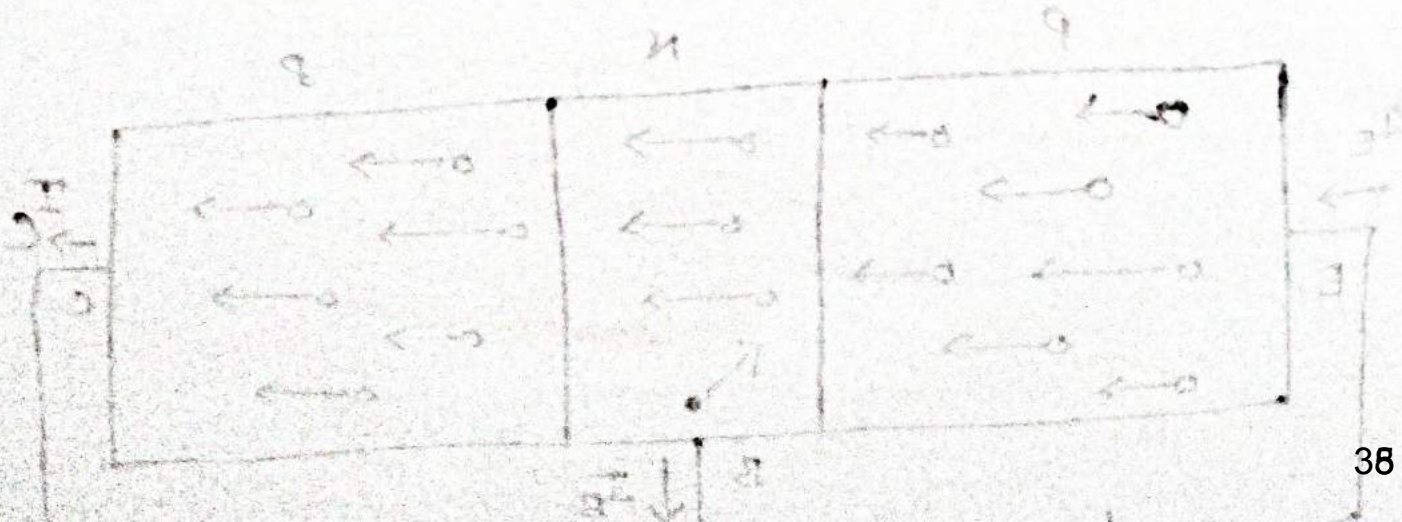
$I_C \rightarrow$ Collector Current.

✓ Emitter Current is sum of Base Current & Collector Current.

$$I_E = I_B + I_C$$

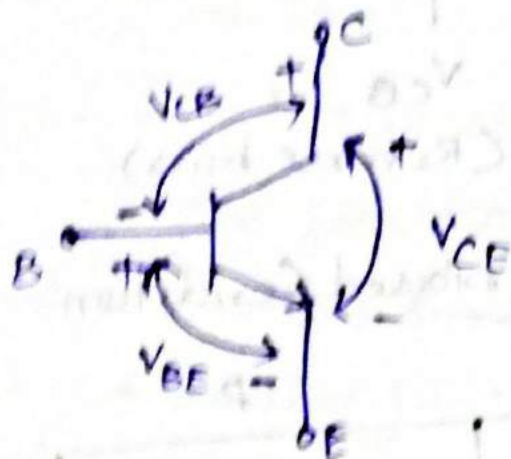


BJT transistor under biased condition

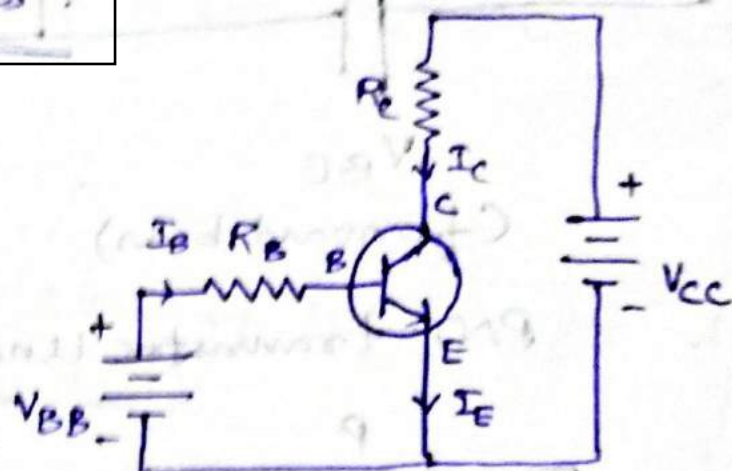


BJT current & voltages :

NPN



(a) Terminal voltage Polarities



(b) Voltage source Connections

$V_{BE} \rightarrow$ Voltage between base & emitter terminals

$V_{BE} \rightarrow$ Cut-in voltage (knee voltage)

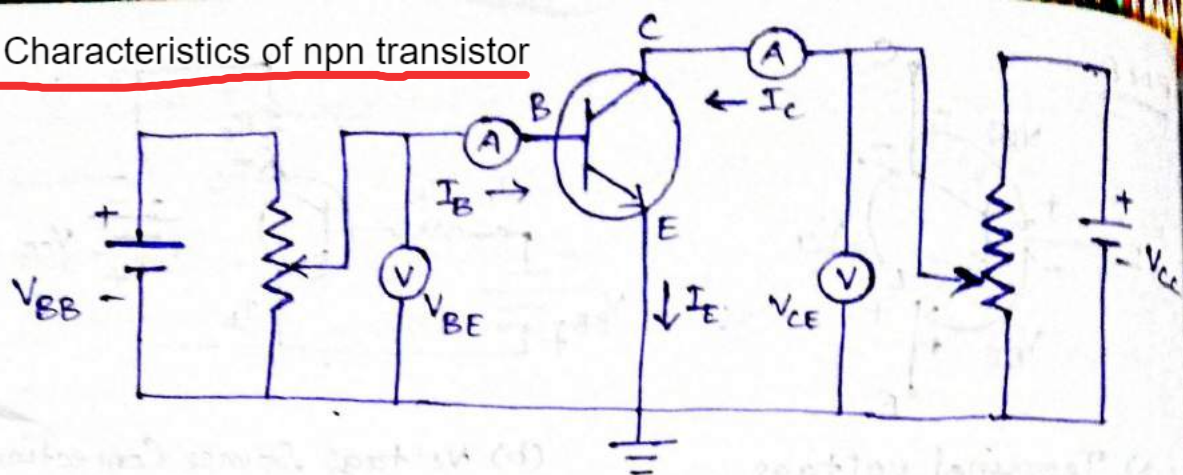
$\rightarrow 0.7 \text{ V}$ for silicon

$\rightarrow 0.3 \text{ V}$ for Germanium

$V_{CB} \rightarrow$ Voltage between collector and base terminals.

$V_{CC} > V_{BB}$ to get a proper conduction due to electrons.

VI Characteristics of npn transistor

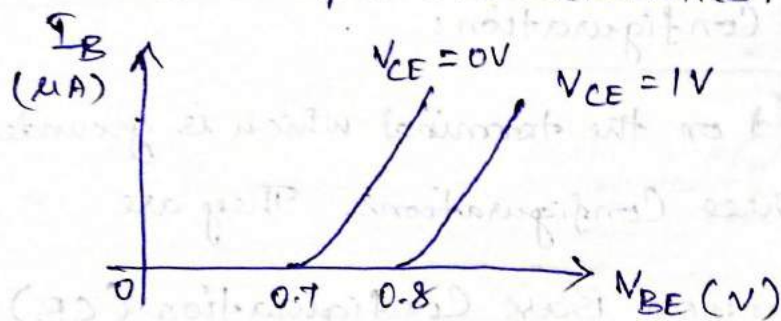


Common Emitter Configured Circuit diagram

Input characteristics:-

✓ The relation between the input current (I_B) and input voltage (V_{BE}) when Output voltage (V_{CE}) is kept constant.

✓ This is input characteristics.



input characteristics

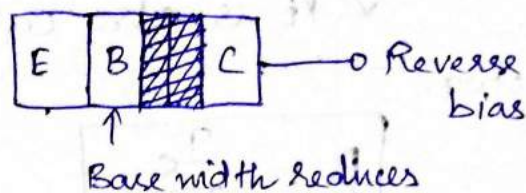
✓ When Output voltage (V_{CE}) is zero ($0V$), the Base Emitter junction behave as a forward biased PN Junction diode.

✓ When Output voltage (V_{CE}) is increased, the Base current Conduction starts ^{later} ~~earlier~~ than at $0V$.

✓ I_B decreases with the increase of output voltage V_{CE} .

✓ This is because of Base width modulation.

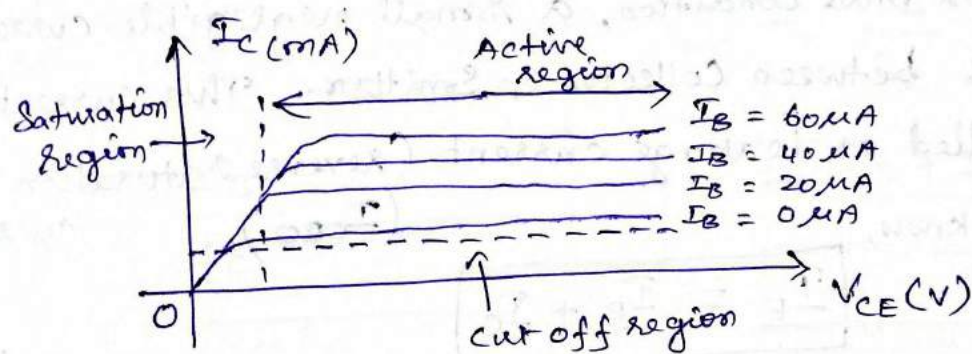
Base width modulation: When reverse voltage V_{CE} increases, the Base width reduces. This is called Base width modulation.



Output characteristics :-

✓ The Relation between the Output current (I_C) and Output voltage (V_{CE}) when input current (I_B) is kept Constant.

✓ This is Output characteristics.



Output characteristics

✓ When Input current (I_B) is Zero ($0\mu A$), The Output voltage (V_{CE}) is increased in uniform steps and output current (I_C) is noted.

✓ This is repeated for different values of Input current (I_B). (Such as. $0\mu A$, $20\mu A$, $40\mu A$, $60\mu A$)

- In Saturation region, it acts as an ON switch
- In Cutoff region, it acts as an OFF switch
- In Active region, it can be used as an Amplifier

Write the expression for α_{dc} and β_{dc} . Derive the relationship between them.

Current gain :-

✓ It is ^{the} ratio between the Output current (I_c) and Input current (I_b).

✓ It is denoted as β

✓ It is also called as Current amplification factor.

$$\beta = \frac{I_c}{I_b}$$

✓ β is greater than ~~or equal to~~ 1.



Relationship between α & β : $\therefore \left[1 + \beta = \frac{1}{1 - \alpha} \right]$

$\alpha \rightarrow$ Current gain for CB Configured Amplifier

$\beta \rightarrow$ Current gain for CE Configured Amplifier

$$\boxed{\beta = \frac{I_C}{I_B}}$$

$$\frac{1}{\beta} = \frac{I_B}{I_C} \therefore \boxed{I_E = I_B + I_C}$$

$$\begin{aligned} \frac{1}{\beta} &= \frac{I_B}{I_C} \\ &= \frac{I_E - I_C}{I_C} \end{aligned}$$

$$= \frac{I_E}{I_C} - \frac{I_C}{I_C}$$

$$= \frac{1}{\alpha} - 1$$

$$\frac{1}{\beta} = \frac{1 - \alpha}{\alpha}$$

$$\boxed{\beta = \frac{\alpha}{1 - \alpha}}$$

$$\boxed{\alpha = \frac{I_C}{I_E}}$$

$$\frac{1}{\alpha} = \frac{I_E}{I_C}$$

$$= \frac{I_B + I_C}{I_C}$$

$$= \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

$$= \frac{1}{\beta} + 1$$

$$\frac{1}{\alpha} = \frac{1 + \beta}{\beta}$$

$$\boxed{\alpha = \frac{\beta}{1 + \beta}}$$

* Problems:

- ① calculate I_c and I_E for a transistor that has $\alpha_{dc} = 0.98$ and $I_B = 100 \mu A$. Determine the value of β_{dc} for the transistor.

Ans:

Given,

$$\alpha = 0.98$$

$$I_B = 100 \mu A$$

We know,

$$\beta = \frac{I_c}{I_B} \quad \text{and} \quad \beta = \frac{\alpha}{1-\alpha}$$

So,

$$\beta = \frac{0.98}{1-0.98}$$

$$\boxed{\beta = 49}$$

$$I_c = \beta I_B$$

$$= 49 \times 100 \times 10^{-6}$$

$$\boxed{I_c = 4.9 \text{ mA}}$$

We know,

$$I_E = I_B + I_c$$

$$= 100 \times 10^{-6} + 4.9 \times 10^{-3}$$

$$\boxed{I_E = 5 \text{ mA}}$$

- ② A transistor has an emitter current of 8 mA , $\alpha = 0.99$. Calculate the collector current.

Ans:

Given,

$$I_E = 8 \text{ mA}$$

$$\alpha = 0.99$$

We know,

$$\alpha = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E = 0.99 \times 8 \times 10^{-3}$$

$$\boxed{I_C = 7.92 \text{ mA}}$$

③ Calculate α_{dc} and β_{dc} for the transistor if I_C is measured as 1 mA & I_B is $25 \mu\text{A}$. Also determine the new base current to give I_C is 5 mA .

Ans:

Given, $I_C = 1 \text{ mA}$, $I_B = 25 \mu\text{A}$

$$\alpha = \frac{I_C}{I_E} = \frac{\beta}{1+\beta}$$
$$= \frac{40}{1+40}$$

$$\beta = \frac{I_C}{I_B}$$
$$= \frac{1 \times 10^{-3}}{25 \times 10^{-6}}$$

$$\boxed{\alpha = \frac{40}{41} = 0.9756}$$

$$\boxed{\beta = 40}$$

Given, when $I_C = 5 \text{ mA}$,

$$\beta = \frac{I_C}{I_B}$$

$$\boxed{I_B = \frac{I_C}{\beta} = \frac{5 \times 10^{-3}}{40} = 125 \mu\text{A}}$$

⑤ A transistor Amplifier Connected in CE mode $h_{fe} = 100$ and $I_B = 50 \mu A$. Compute the values of I_C and α .

Ans:

Given, $h_{fe} = \beta = 100$
 $I_B = 50 \mu A$

$$\alpha = \frac{\beta}{1 + \beta} = \frac{100}{101} = 0.99$$

$$\boxed{\alpha = 0.99}$$

$$\alpha = \frac{I_c}{I_e}$$

$$\beta = \frac{I_c}{I_B}$$

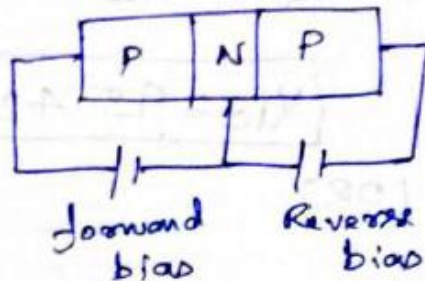
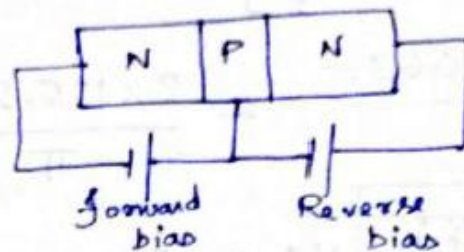
$$I_c = \beta I_B$$

$$= 100 \times 50 \times 10^{-6}$$

$$I_c = 5 \text{ mA}$$

BJT Biasing :-

✓ Applying voltage sources in the input side and Output side properly for the purpose of Amplification. This Applying voltage sources is called BJT Biasing.

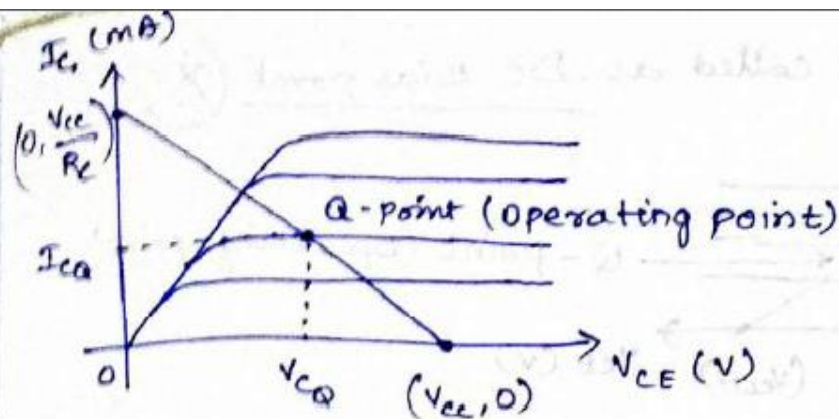


- Analyze the DC load line of CE amplifier and its significance

DC Load line :

- ✓ A straight line drawn on the transistor Output characteristics which is used to operate the BJT in the proper Amplification mode.
- ✓ A straight line drawn between the points taken on x-axis (when $I_C = 0 \text{ mA}$) and taken on y-axis (when $V_{CE} = 0 \text{ V}$).

This is called DC load line.



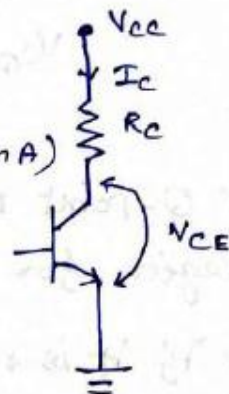
when $I_C = 0 \text{ mA}$, will get the point on x-axis,

$$V_{CC} = I_C R_C + V_{CE}$$

$$= 0 + V_{CE} \quad \because (I_C = 0 \text{ mA})$$

$$\boxed{V_{CC} = V_{CE}}$$

* Coordinates on x-axis — $(V_{CC}, 0)$



When $V_{CE} = 0 \text{ V}$, will get the point on y-axis,

$$V_{CC} = I_C R_C + V_{CE}$$

$$V_{CC} = I_C R_C + 0$$

$$\boxed{I_C = \frac{V_{CC}}{R_C}}$$

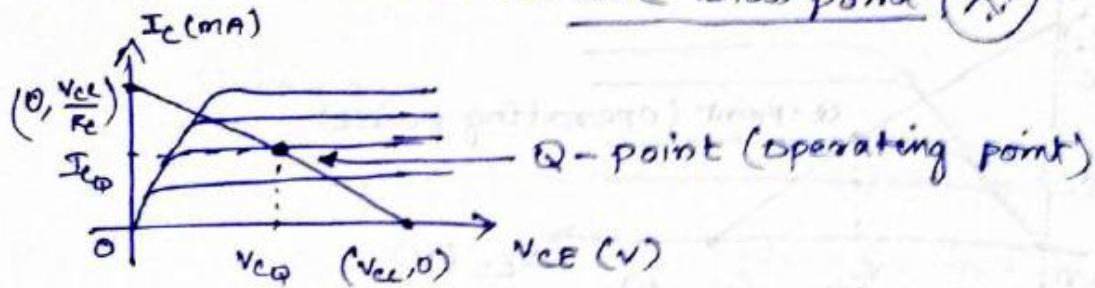
* Coordinates on y-axis — $(0, \frac{V_{CC}}{R_C})$

A line connecting the * & * coordinates on x-axis & y-axis called DC load line.

Q-point (Quiescent point) (Operating point)

✓ it is point chosen on the intersection between the DC load line & Output characteristics of the Transistor

✓ It is also called as DC Bias point (X)



where, $I_{CQ} \rightarrow$ Collector current at Q-point (I_{CQ})

$V_{CEQ} \rightarrow$ Collector Emitter Voltage (V_{CEQ}) at Q-point

✓ Q-point must be fixed at the center of Active region for faithful amplification.

✓ if it is not fixed, distortion will arise in the Output.

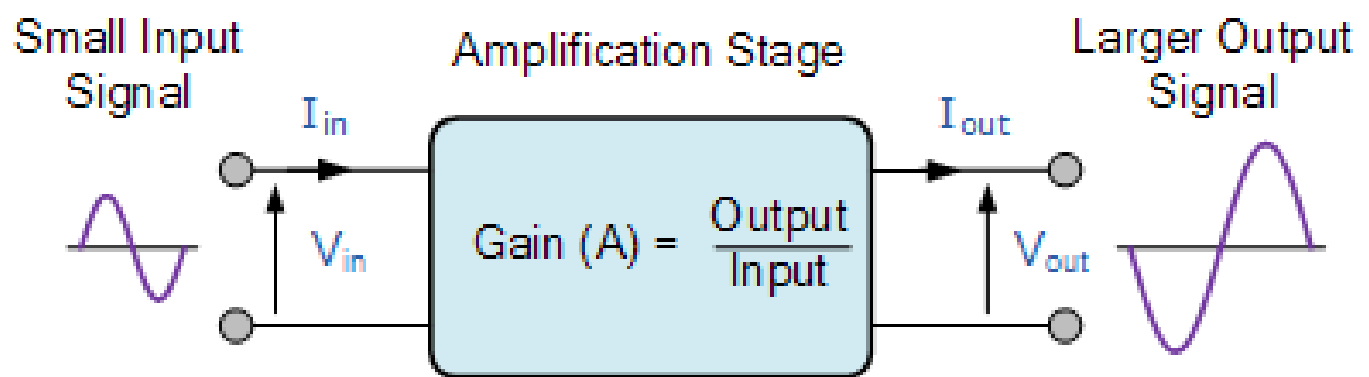
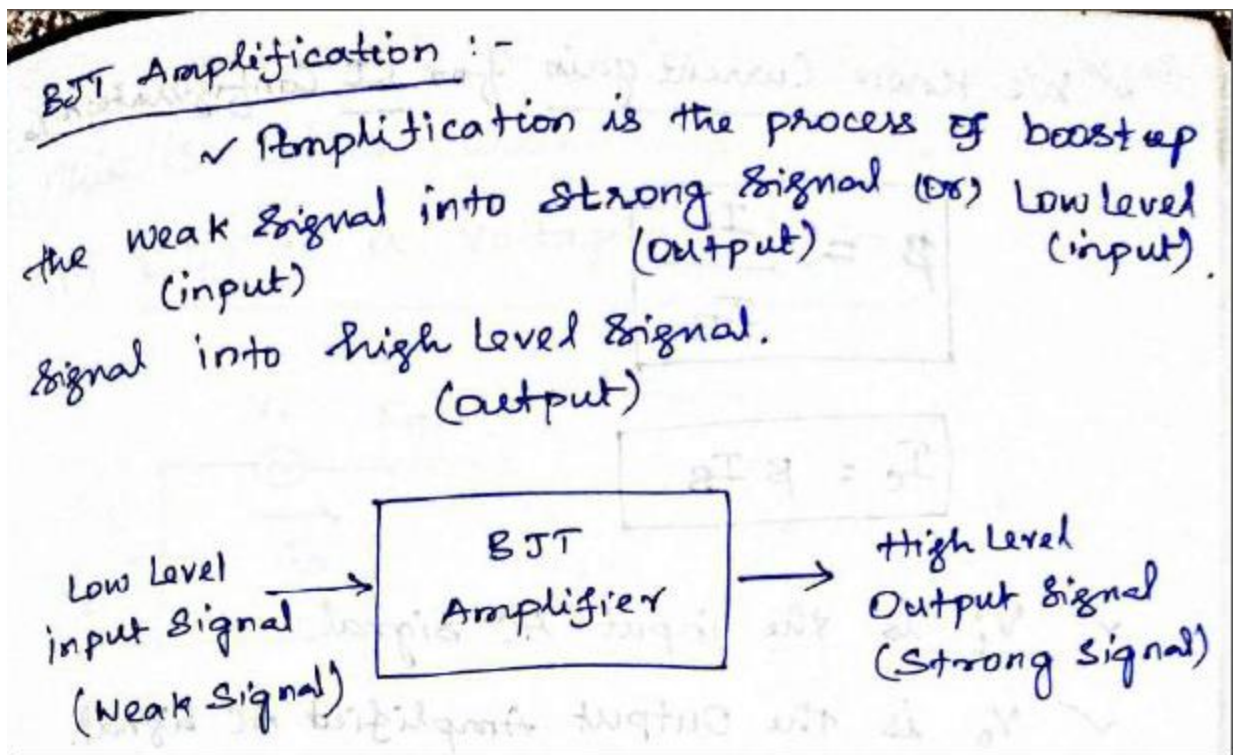
Need for Biasing

✓ For faithful amplification, the operating point (Q-point) must be at the center of the active region

✓ To operate the transistor in the desired region, external DC voltage of correct polarity & magnitude is applied.

Conditions for proper biasing

- 1: Proper DC value of Collector current (I_C)
- 2: Proper value of V_{BE} (0.7V for Si, 0.3V for Ge)
- 3: Proper value of V_{CE} (1V for Si, 0.5V for Ge)



dissipation. The product of V_{CE} and I_C must not exceed the maximum power dissipation, $P_{D(max)}$. (Data sheets may show this as simply P_D in a column labeled Maximum Ratings.) Both V_{CE} and I_C cannot be maximum at the same time. If V_{CE} is maximum, I_C can be calculated as

$$I_C = \frac{P_{D(max)}}{V_{CE}}$$

If I_C is maximum, V_{CE} can be calculated by rearranging the previous equation as follows:

$$V_{CE} = \frac{P_{D(max)}}{I_C}$$

The BJT as an Amplifier

Amplification is the process of linearly increasing the amplitude of an electrical signal and is one of the major properties of a transistor.

VOLTAGE AMPLIFICATION

The transistor amplifies current because the collector current is equal to the base current multiplied by the current gain, β . The base current in a transistor is very small compared to the collector and emitter currents. Because of this, the collector current is approximately equal to the emitter current. With this in mind, let's look at the circuit in Figure 2–21. An ac voltage, V_s , is superimposed on the dc bias voltage V_{BB} by capacitive coupling as shown. The dc bias voltage V_{CC} is connected to the collector through the collector resistor, R_C

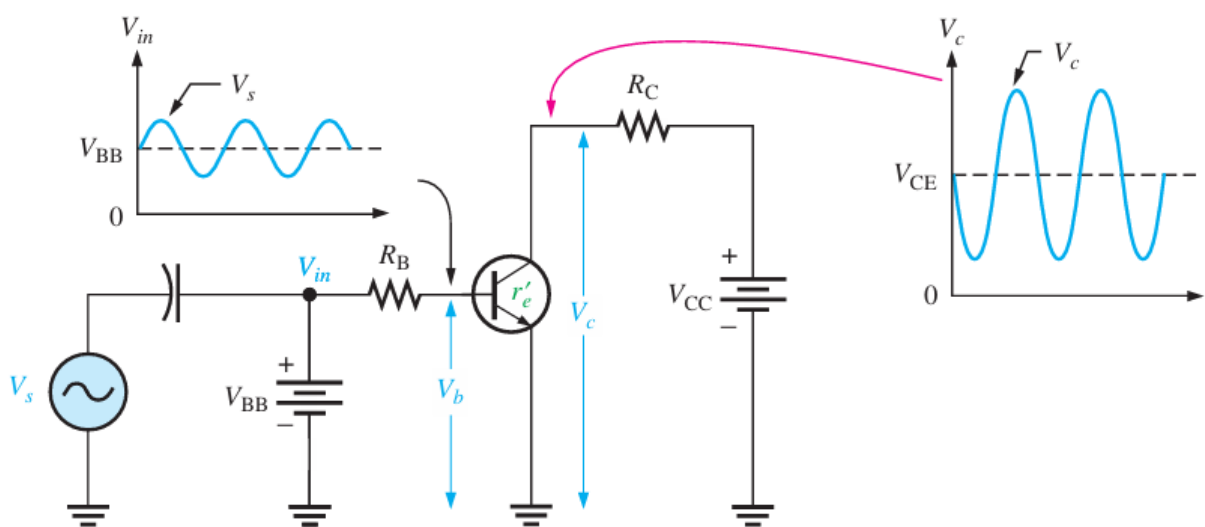


FIGURE 2–21 Basic transistor amplifier circuit

The ac input voltage produces an ac base current, which results in a much larger ac collector current. This ac collector current produces an ac voltage across R_C , thus producing an amplified, but inverted, reproduction of the ac input voltage in the active region of operation, as illustrated in Figure 2–21. The forward-biased base-emitter junction presents a very low resistance to the ac signal. This internal ac emitter resistance is designated r'_e in Figure 2–21 and appears in series with R_B . The ac base voltage is

$$V_b = I_e r'_e$$

The ac collector voltage, V_c , equals the ac voltage drop across R_C

$$V_c = I_c R_C$$

Since $I_c = I_e$, the ac collector voltage is

$$V_c \cong I_e R_C$$

V_b can be considered the transistor ac input voltage where $V_b = V_s - I_b R_B$. V_c can be considered the transistor ac output voltage. Since voltage gain is defined as the ratio of the output voltage to the input voltage, the ratio of V_c to V_b is the ac voltage gain, A_v , of the transistor

$$A_v = \frac{V_c}{V_b}$$

$$A_v = \frac{V_c}{V_b} \cong \frac{I_e R_C}{I_e r'_e}$$

$$A_v \cong \frac{R_C}{r'_e}$$

THE BJT AS A SWITCH

Switching Operation Figure 2–23 illustrates the basic operation of a BJT as a switching device. In part (a), the transistor is in the cutoff region because the base-emitter junction is not forward-biased. In this condition, there is, ideally, an open between collector and emitter, as indicated by the switch equivalent. In part (b), the transistor is in the saturation region because the base-emitter junction and the base-collector junction are forward-biased and the base current is made

large enough to cause the collector current to reach its saturation value. In this condition, there is, ideally, a short between collector and emitter, as indicated by the switch equivalent. Actually, a small voltage drop across the transistor of up to a few tenths of a volt normally occurs, which is the saturation voltage, $V_{CE(sat)}$

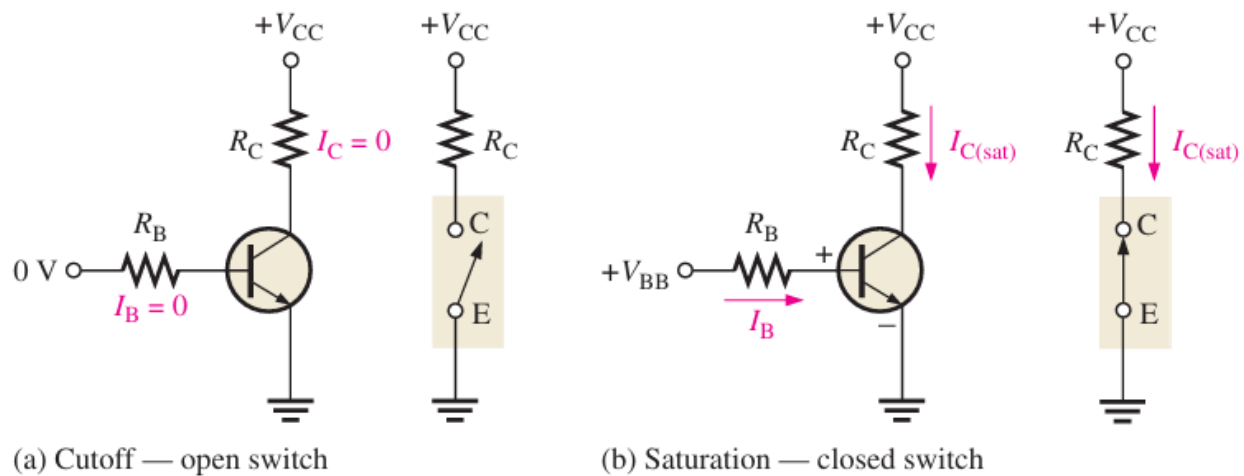


FIGURE 2–23 Switching action of an ideal transistor.

Conditions in Cutoff

As mentioned before, a transistor is in the cutoff region when the base-emitter junction is not forward-biased. Neglecting leakage current, all of the currents are zero, and V_{CE} is equal to V_{CC} .

$$V_{CE(cutoff)} = V_{CC}$$

Conditions in Saturation

As you have learned, when the base-emitter junction is forward-biased and there is enough base current to produce a maximum collector current, the transistor is saturated. The formula for collector saturation current is

$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C}$$

Since $V_{CE(sat)}$ is very small compared to V_{CC} , it can usually be neglected. The minimum value of base current needed to produce saturation is

$$I_{B(min)} = \frac{I_{C(sat)}}{\beta_{DC}}$$

Normally, I_B should be significantly greater than $I_{B(min)}$ to ensure that the transistor is saturated.

A Simple Application of a Transistor Switch

The transistor in Figure 2–25 can be used as a switch to turn the LED on and off. For example, a square wave input voltage with a period of 2 s is applied to the input as indicated. When the square wave is at 0 V, the transistor is in cutoff; and since there is no collector current, the LED does not emit light. When the square wave goes to its high level and there is sufficient base current, the transistor saturates. This forward-biases the LED, and the resulting collector current through the LED causes it to emit light. Thus, the LED is on for 1 second and off for 1 second. The following example illustrates the switching operation for given circuit values.

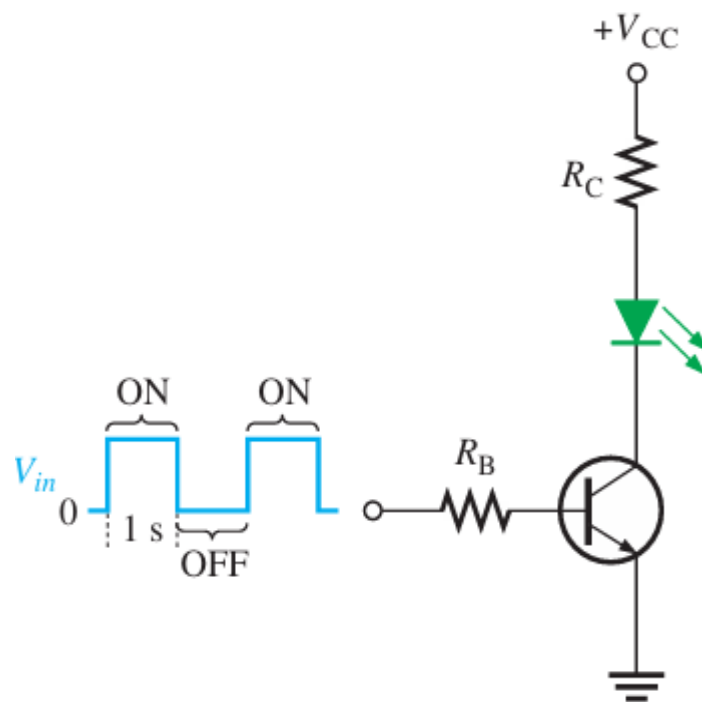


FIGURE 2–25 A transistor used to switch an LED on and off with proper R values selected based on transistor parameters.

Need for Biasing

Transistor must be properly biased in order to operate as an amplifier. DC biasing is used to establish fixed dc values for the transistor currents and voltages called the dc operating point or quiescent point (Q-point).

The DC Operating Point

A transistor must be properly biased with a dc voltage in order to operate as a linear amplifier. A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.

Linear Operation

The region along the load line including all points between saturation and cutoff is generally known as the linear region of the transistor's operation. As long as the transistor is operated in this region, the output voltage is ideally a linear reproduction of the input.

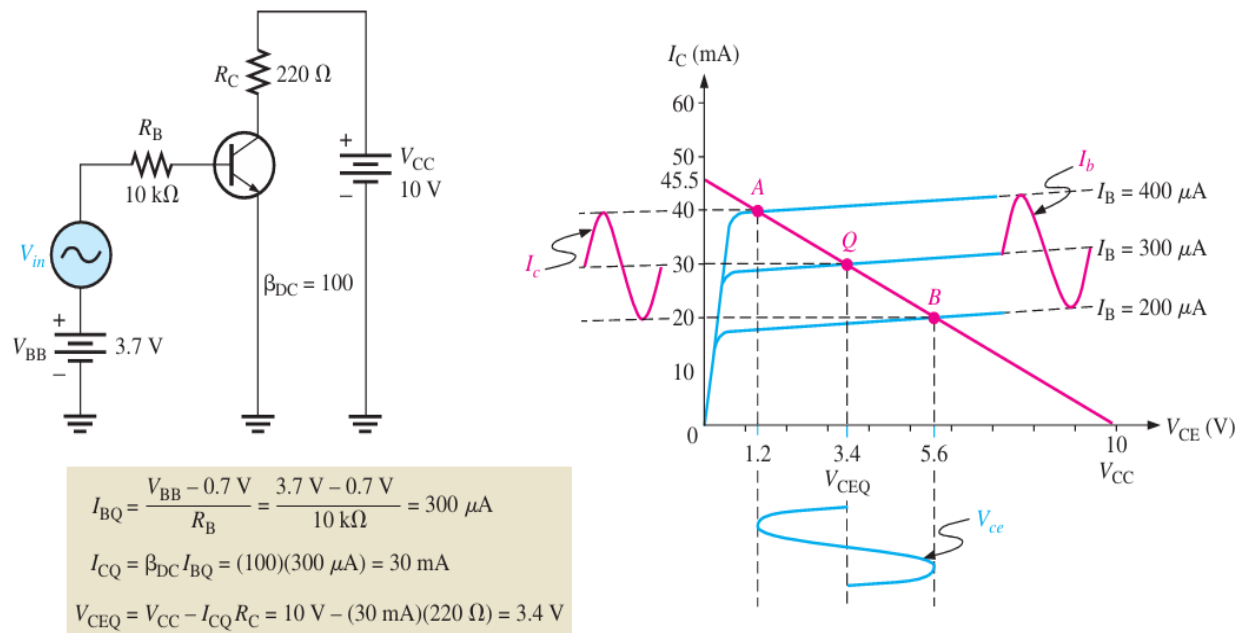


FIGURE : Variations in collector current and collector-to-emitter voltage as a result of a variation in base current

The JFET:

The JFET (junction field-effect transistor) is a type of FET that operates with a reverse-biased pn junction to control current in a channel. Depending on their structure, JFETs fall into either of two categories, n channel or p channel.

Basic Structure:

Figure (a) shows the basic structure of an n-channel JFET (junction field-effect transistor). Wire leads are connected to each end of the n-channel; the drain is at the upper end, and the source is

at the lower end. Two p-type regions are diffused in the n-type material to form a channel, and both p-type regions are connected to the gate lead. The gate lead is shown connected to only one of the p regions, which are internally connected together. A p-channel JFET is shown in Figure (b).

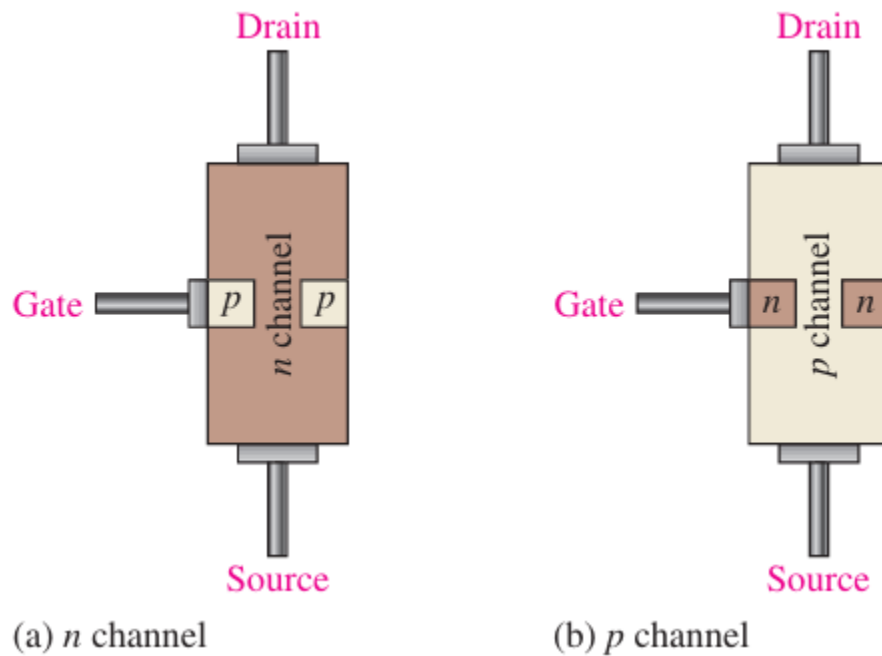


FIGURE: A

representation of the basic structure of the two types of JFET.

Basic Operation

To illustrate the operation of a JFET, Figure shows dc bias voltages applied to an n-channel device. V_{DD} provides a drain-to-source voltage and supplies current from

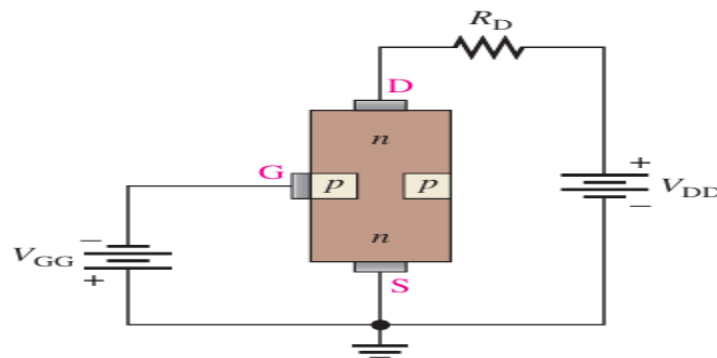


Figure: A biased n-channel JFET

drain to source. V_{GG} sets the reverse-bias voltage between the gate and the source, as shown. The JFET is always operated with the gate-source pn junction reverse-biased. Reverse biasing of the gate-source junction produces a depletion region along the pn junction, which extends into the channel and thus increases its resistance by restricting the channel width. The channel width and thus the channel resistance can be controlled by varying the gate voltage, thereby controlling the amount of drain current, I_D . Above Figure illustrates this concept with an n-channel device. The white areas represent the depletion region created by the reverse bias. It is wider toward the drain end of the channel because the reverse-bias voltage between the gate and the drain is greater than that between the gate and the source.

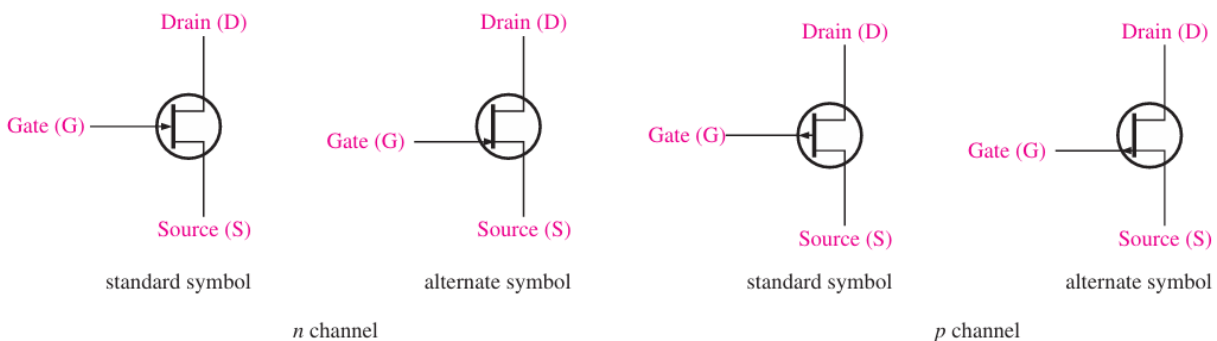


Figure: JFET schematic symbols.

JFET Characteristics

Drain Characteristic Curve Consider the case when the gate-to-source voltage is zero ($V_{GS} = 0$ V). This is produced by shorting the gate to the source, as in Figure (a) where both are grounded. As V_{DD} (and thus V_{DS}) is increased from 0 V, I_D will increase proportionally, as shown in the graph of Figure (b) between points A and B. In this area, the channel resistance is essentially constant because the depletion region is not large enough to have significant effect. This is called the ohmic region because V_{DS} and I_D are related by Ohm's law.

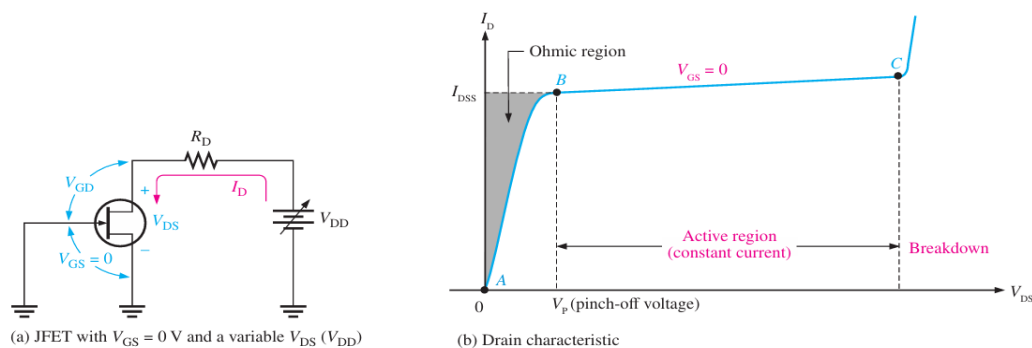


FIGURE (b) Drain characteristic curve of a JFET for $V_{GS} = 0$ showing pinch-off voltage.

At point B in Figure (b), the curve levels off and enters the active region where I_D becomes essentially constant. As V_{DS} increases from point B to point C, the reverse-bias voltage from gate to drain (V_{GD}) produces a depletion region large enough to offset the increase in V_{DS} , thus keeping I_D relatively constant. Pinch-Off Voltage For $V_{GS} = 0$ V, the value of V_{DS} at which I_D becomes essentially constant (point B on the curve in Figure (b)) is the pinch-off voltage, V_P . For a given JFET, V_P has a fixed value. As you can see, a continued increase in V_{DS} above the pinch-off voltage produces an almost constant drain current until point C is reached. This value of drain current is I_{DSS} (Drain to Source current with gate Shorted) and is always specified on JFET datasheets. I_{DSS} is the maximum drain current that a specific JFET can produce regardless of the external circuit, and it is always specified for the condition, $V_{GS} = 0$ V. Breakdown As shown in the graph in Figure (b), breakdown occurs at point C when I_D begins to increase very rapidly with any further increase in V_{DS} . Breakdown can result in irreversible damage to the device, so JFETs are always operated below breakdown and within the active region (constant current)

VGS Controls I_D

Let's connect a reverse-bias voltage, V_{GG} , from gate to source as shown in Figure (a). As V_{GS} is set to increasingly more negative values by adjusting V_{GG} , a family of drain characteristic curves is produced, as shown in Figure (b). Notice that I_D decreases as the magnitude of V_{GS} is increased to larger negative values because of the narrowing of the channel. Also notice that, for each increase in V_{GS} , the JFET reaches pinch-off (where constant current begins) at values of V_{DS} less than V_P . The term pinch-off is not the same as pinch-off voltage, V_P . The JFET must be operated between $V_{GS} = 0$ V and $V_{GS(off)}$. For this range of gate-to-source voltages, I_D will vary from a maximum of I_{DSS} to a minimum of almost zero.

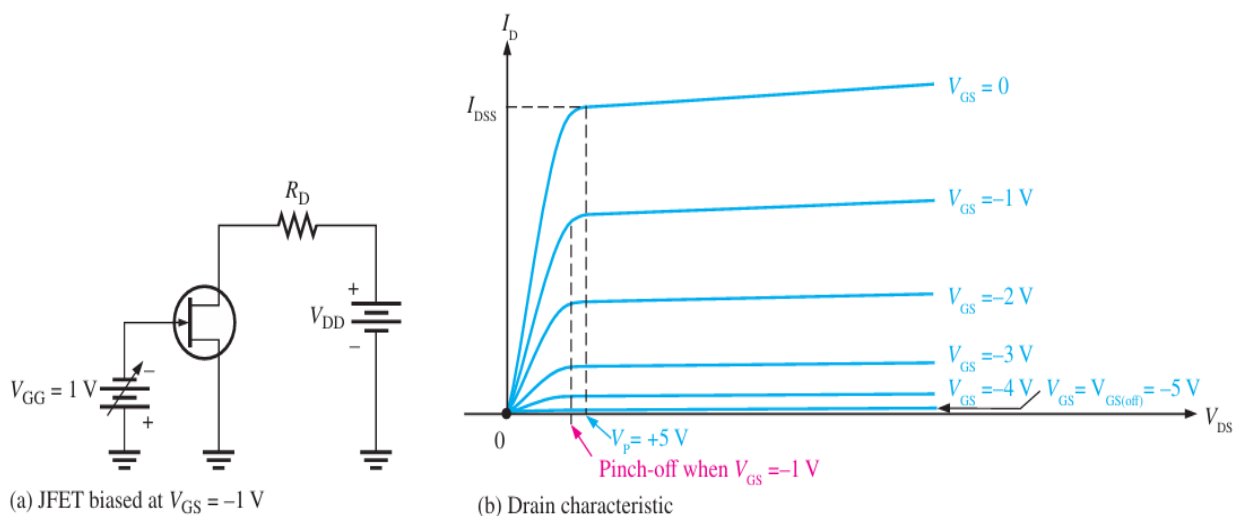


FIGURE: Pinch-off occurs at a lower V_{DS} as V_{GS} is increased to more negative values.

JFET Transfer Characteristics:

You have learned that a range of V_{GS} values from zero to $V_{GS(off)}$ controls the amount of drain current. For an n-channel JFET, $V_{GS(off)}$ is negative, and for a p-channel JFET, $V_{GS(off)}$ is positive. Because V_{GS} does control I_D , the relationship between these two quantities is very important. Figure is a general transfer characteristic curve that illustrates graphically the relationship between V_{GS} and I_D .

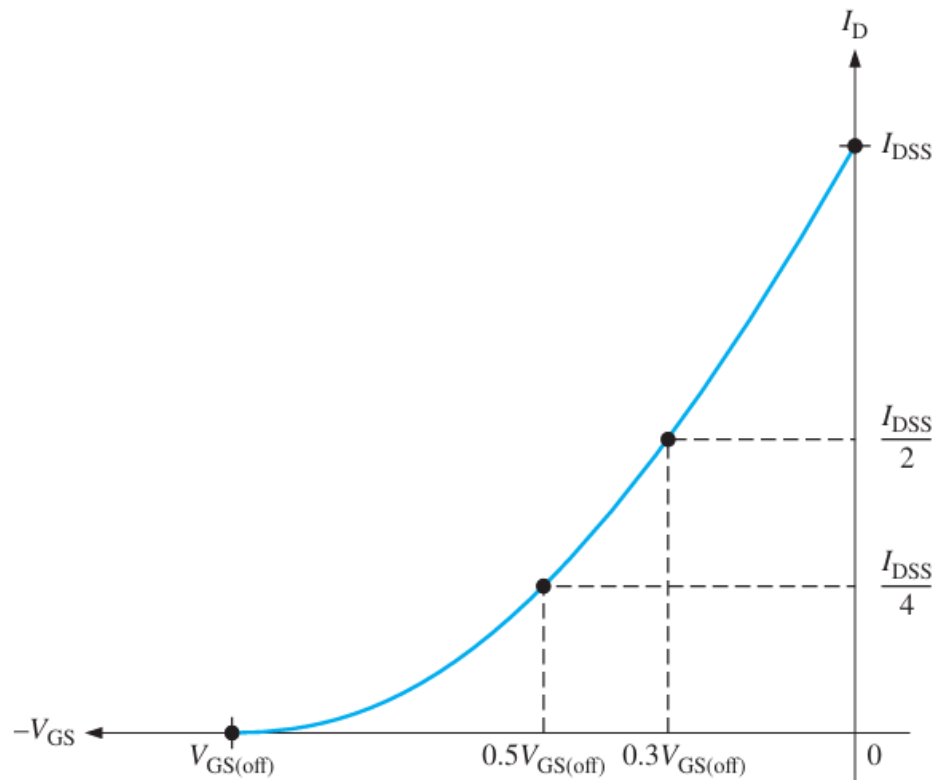


FIGURE: JFET Transfer Characteristic Curve (n-channel).

Notice that the bottom end of the curve is at a point on the V_{GS} axis equal to $V_{GS(off)}$, and the top end of the curve is at a point on the I_D axis equal to I_{DSS} . This curve shows that

$$I_D = 0 \quad \text{when } V_{GS} = V_{GS(off)}$$

$$I_D = \frac{I_{DSS}}{4} \quad \text{when } V_{GS} = 0.5V_{GS(off)}$$

$$I_D = \frac{I_{DSS}}{2} \quad \text{when } V_{GS} = 0.3V_{GS(off)}$$

$$I_D = I_{DSS} \quad \text{when } V_{GS} = 0$$

The transfer characteristic curve can also be developed from the drain characteristic curves by plotting values of I_D for the values of V_{GS} taken from the family of drain curves at pinch-off, as illustrated in Figure for a specific set of curves. Each point on the transfer characteristic curve corresponds to specific values of V_{GS} and I_D on the drain curves. For example, when $V_{GS} = -2$ V, $I_D = 4.32$ mA. Also, for this specific JFET, $V_{GS(off)} = -5$ V and $I_{DSS} = 12$ mA.

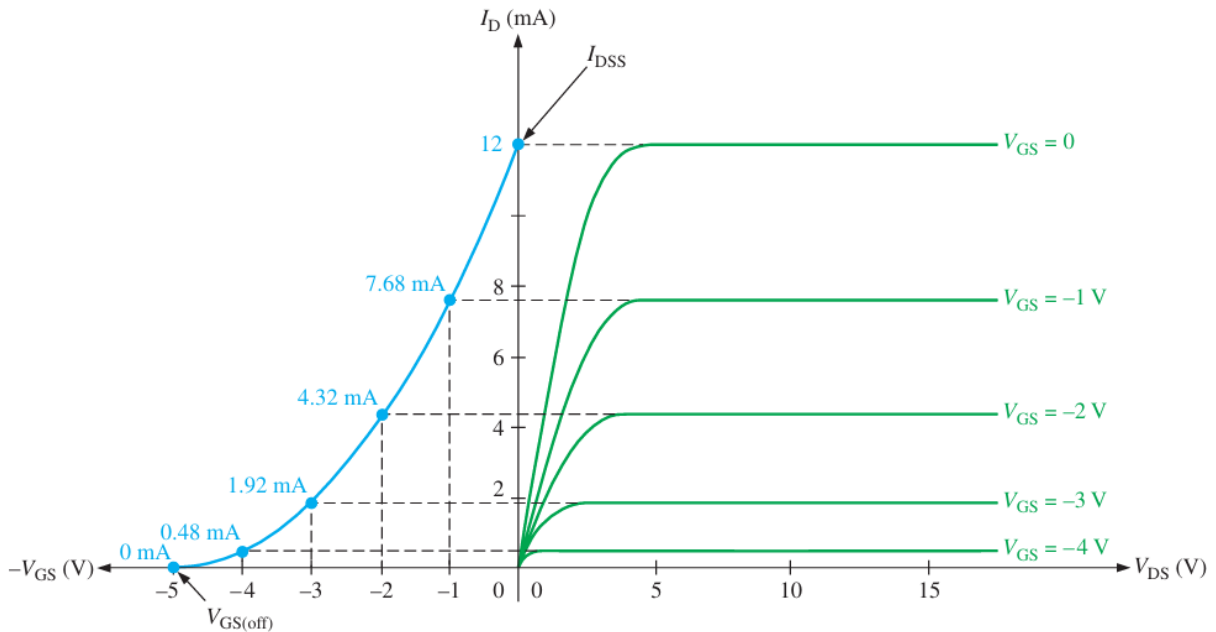


Figure: Example of the development of an n-channel JFET transfer characteristic curve (blue) from the JFET drain characteristic curves (green).

A JFET transfer characteristic curve is expressed approximately as

$$I_D \cong I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2$$